

MAINTENANCE REPORT

IndAI Plant Operations — Confidential

Document No: MR-2024-0478 | Date: 12/07/2024

1. Equipment Details

Field	Details
Equipment ID	P-101
Equipment Type	Centrifugal Pump
Location	Unit 3 — Cooling Circuit
Reported By	Rajesh Kumar (Maintenance Engineer)
Work Order	MO-2478

2. Failure Description

Pump P-101 experienced abnormal vibration on 10/07/2024 during routine inspection. The vibration level exceeded 12 mm/s, surpassing the alarm threshold of 7 mm/s. Operator reported unusual noise from the bearing housing. Immediate shutdown was initiated to prevent further damage. This is the third failure of P-101 in 2024.

3. Root Cause

Inspection revealed severe bearing wear on the drive-end bearing (SKF 6310). Lubrication was found inadequate — grease interval was overdue by 45 days. Possible misalignment between pump and motor shaft also observed (offset: 0.3 mm). Bearing replacement completed. Shaft realigned to within 0.05 mm tolerance.

4. Actions Taken

Bearing replaced with SKF 6310 (new). Lubrication schedule updated — interval reduced to 30 days. Shaft alignment corrected. Vibration post-repair: 2.1 mm/s (within limits). Next inspection scheduled for 12/08/2024.

5. References

OEM Manual Section 5.3 — Bearing Maintenance. ISO 10816-3 Vibration Standard.